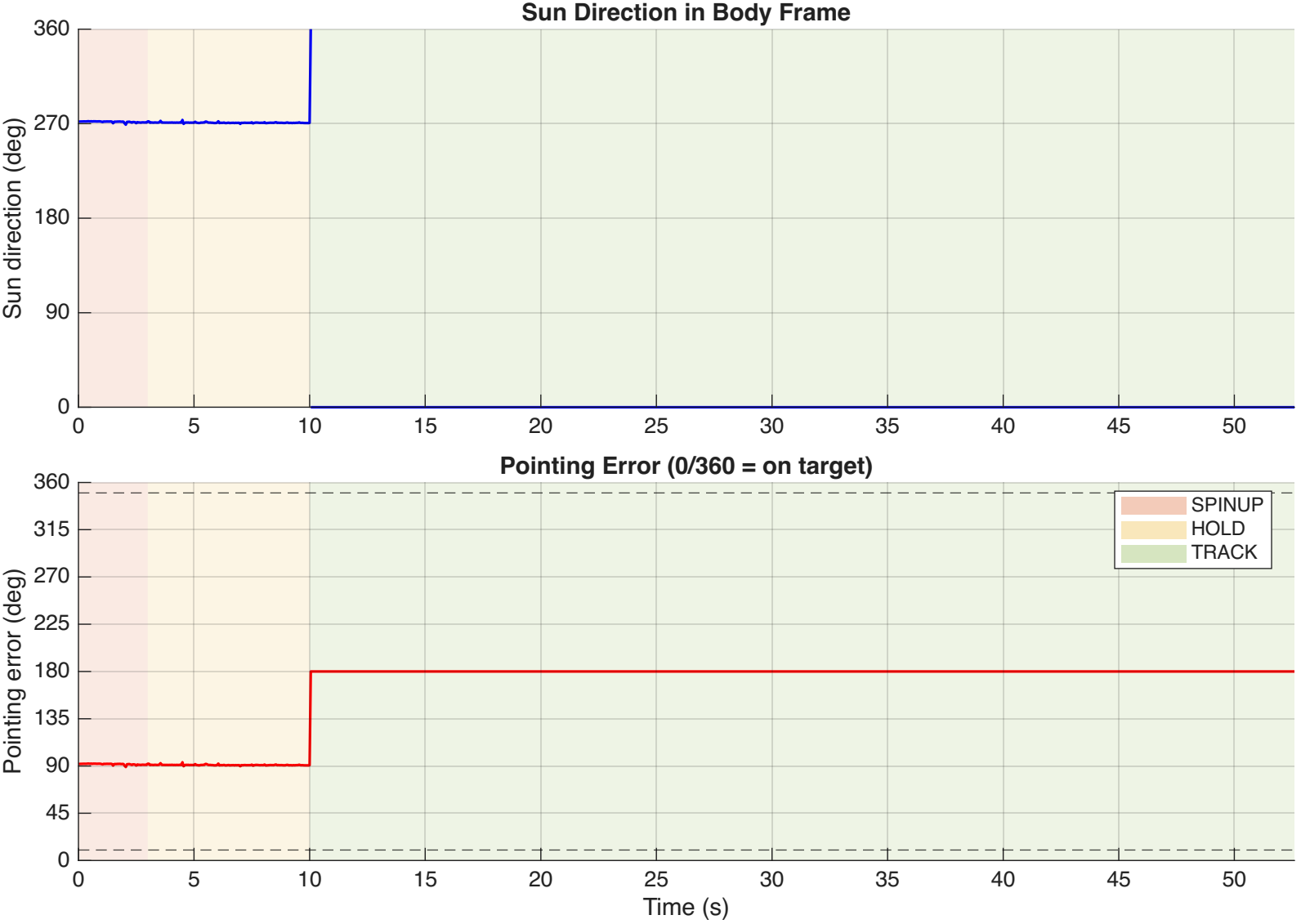
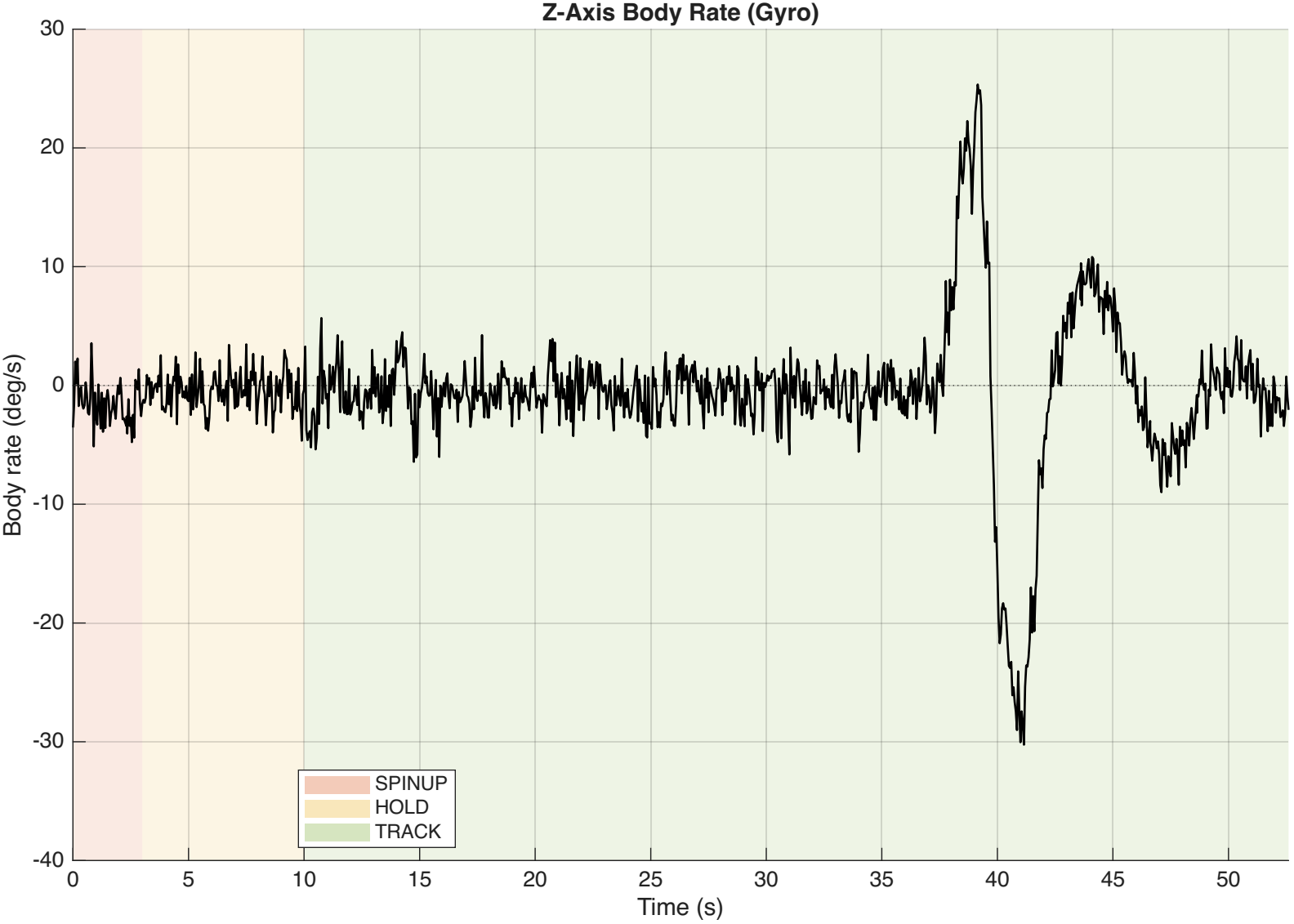
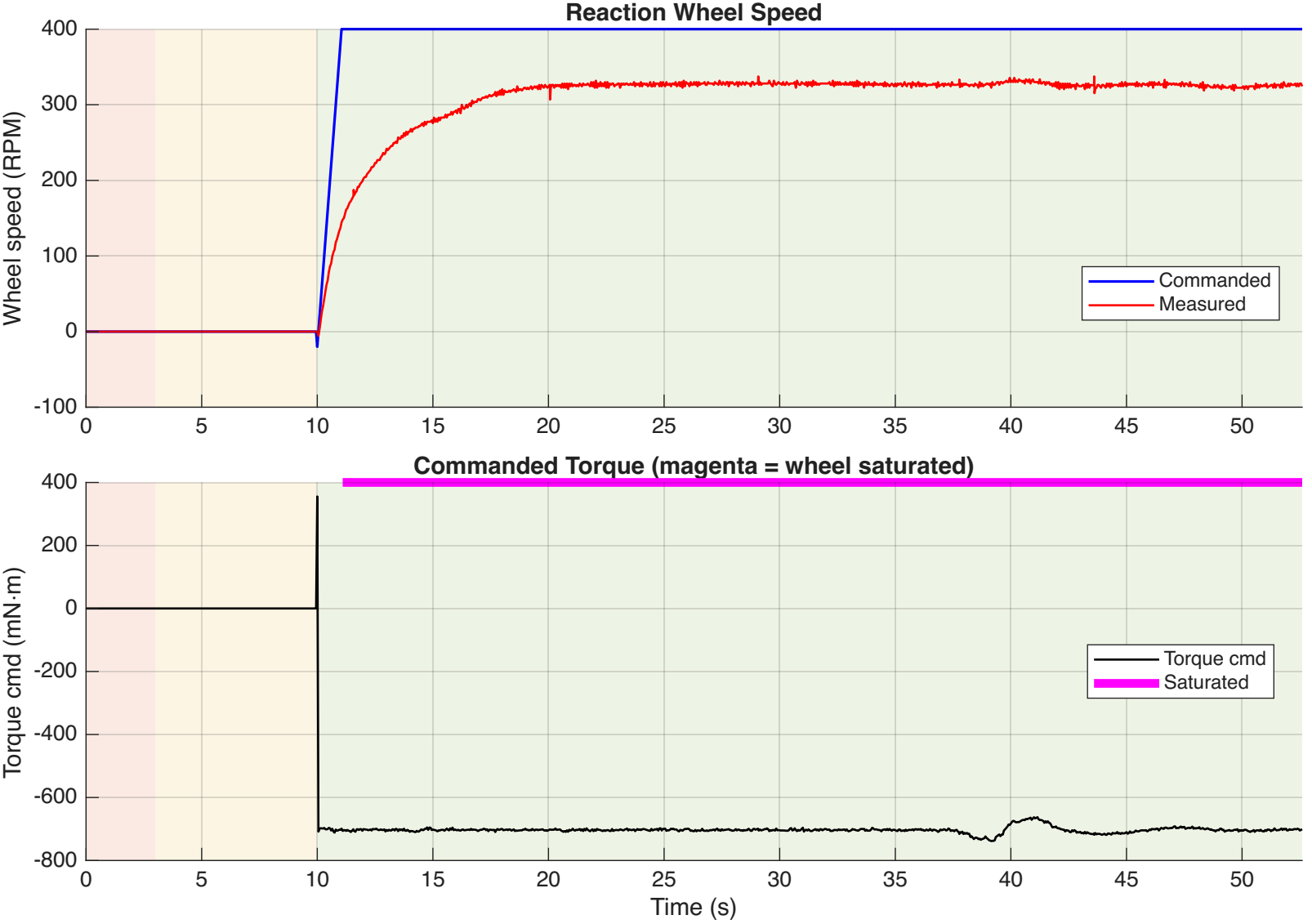
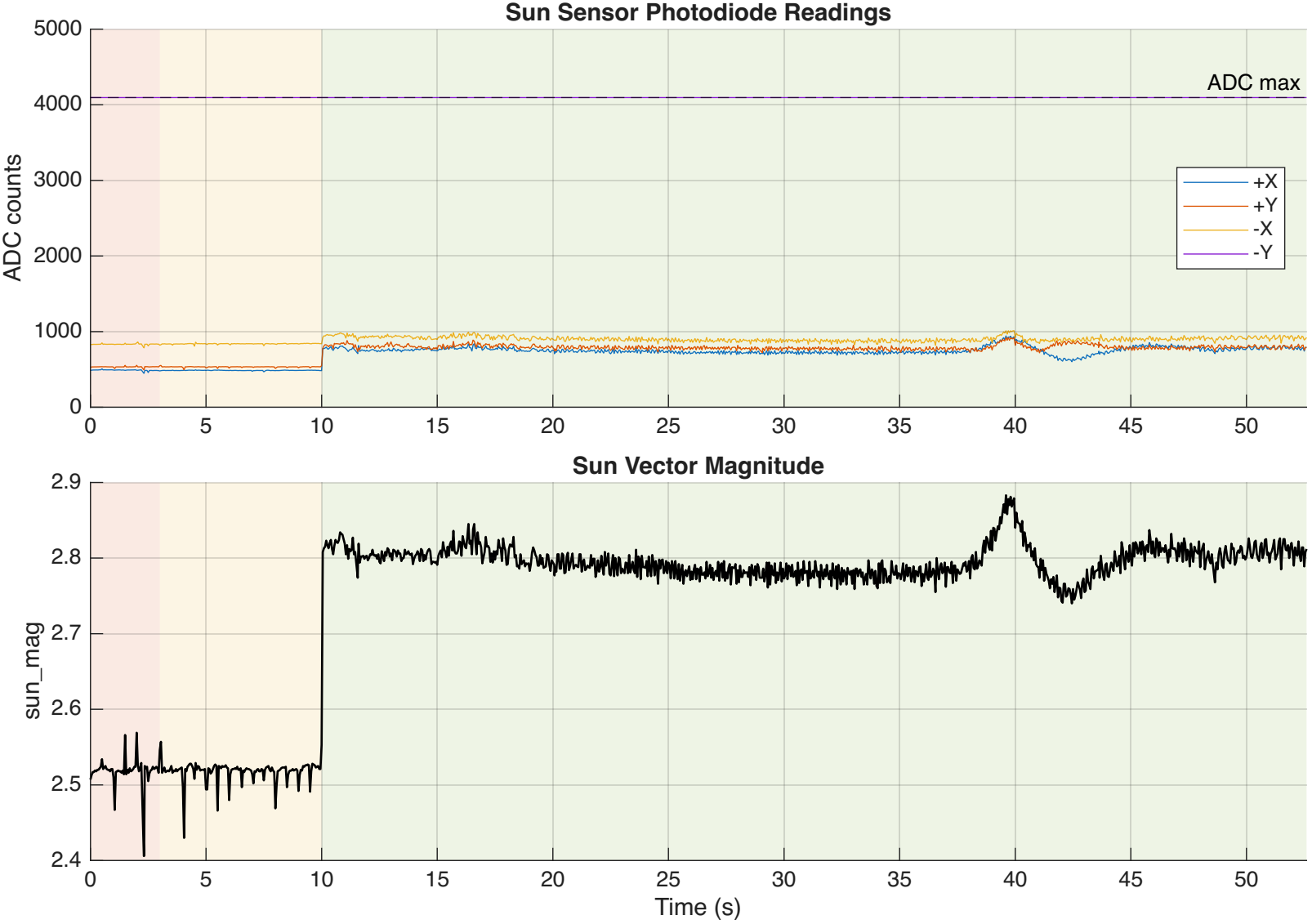
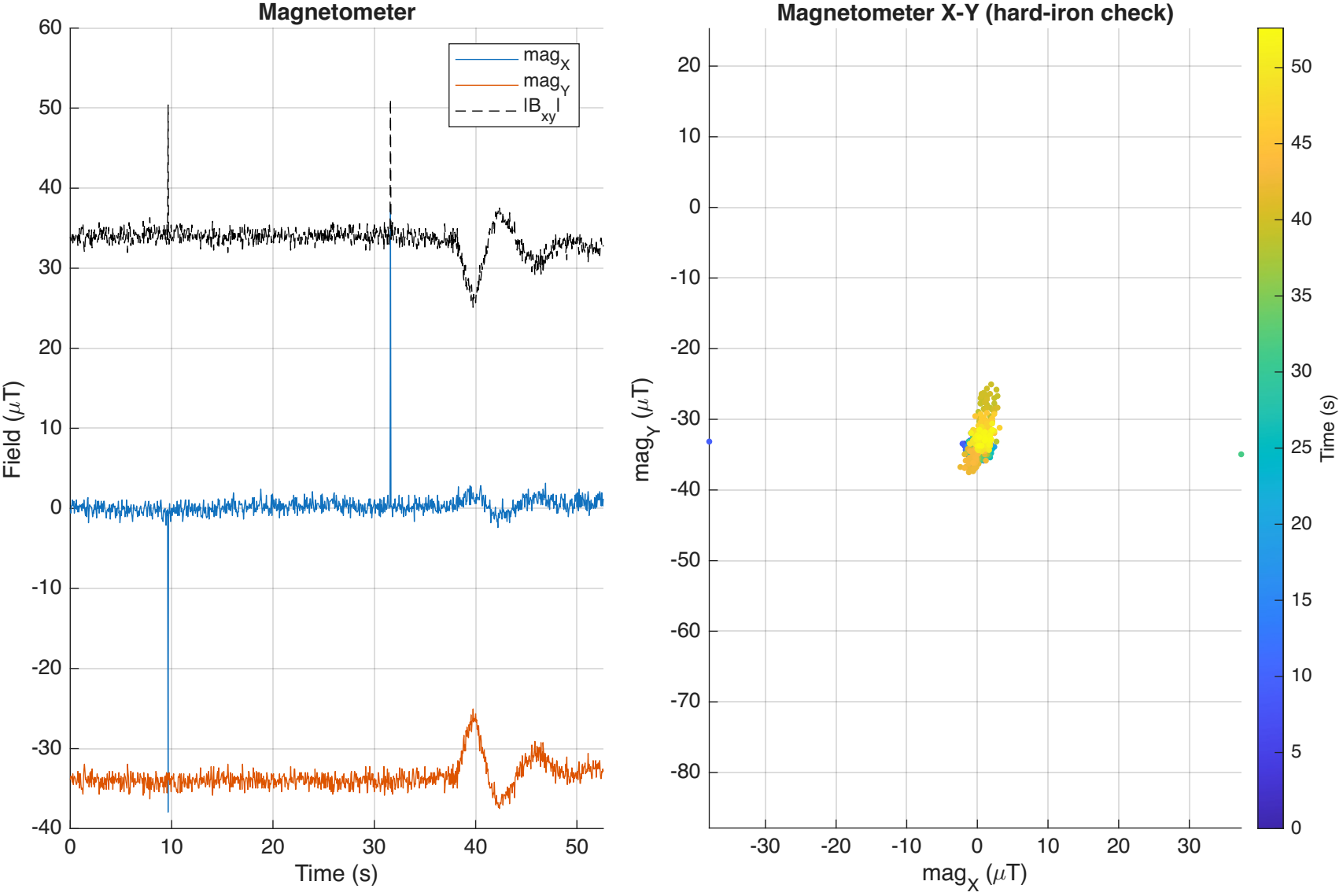
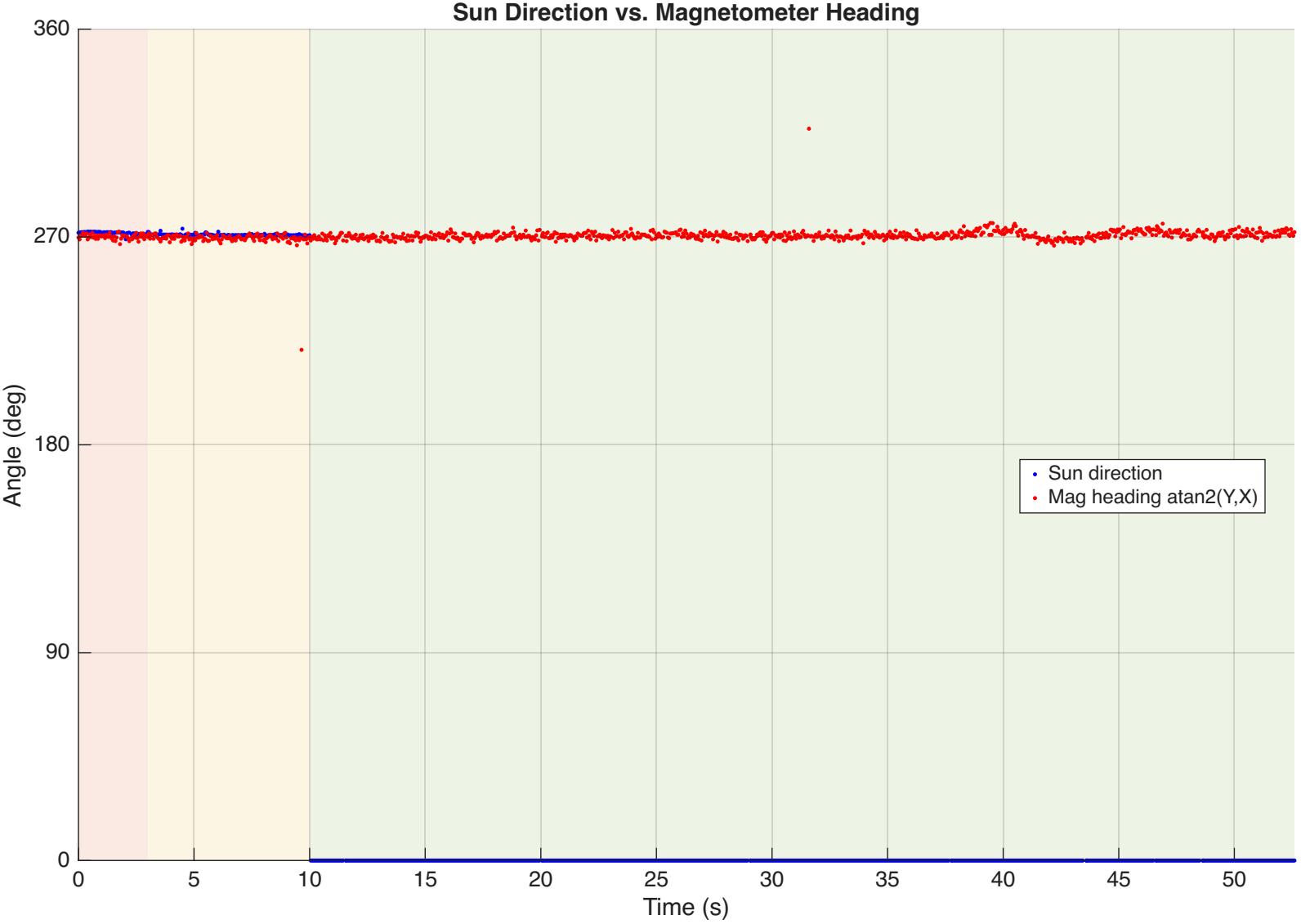

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
5
10
15
20
25
30
35
40
45
50
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

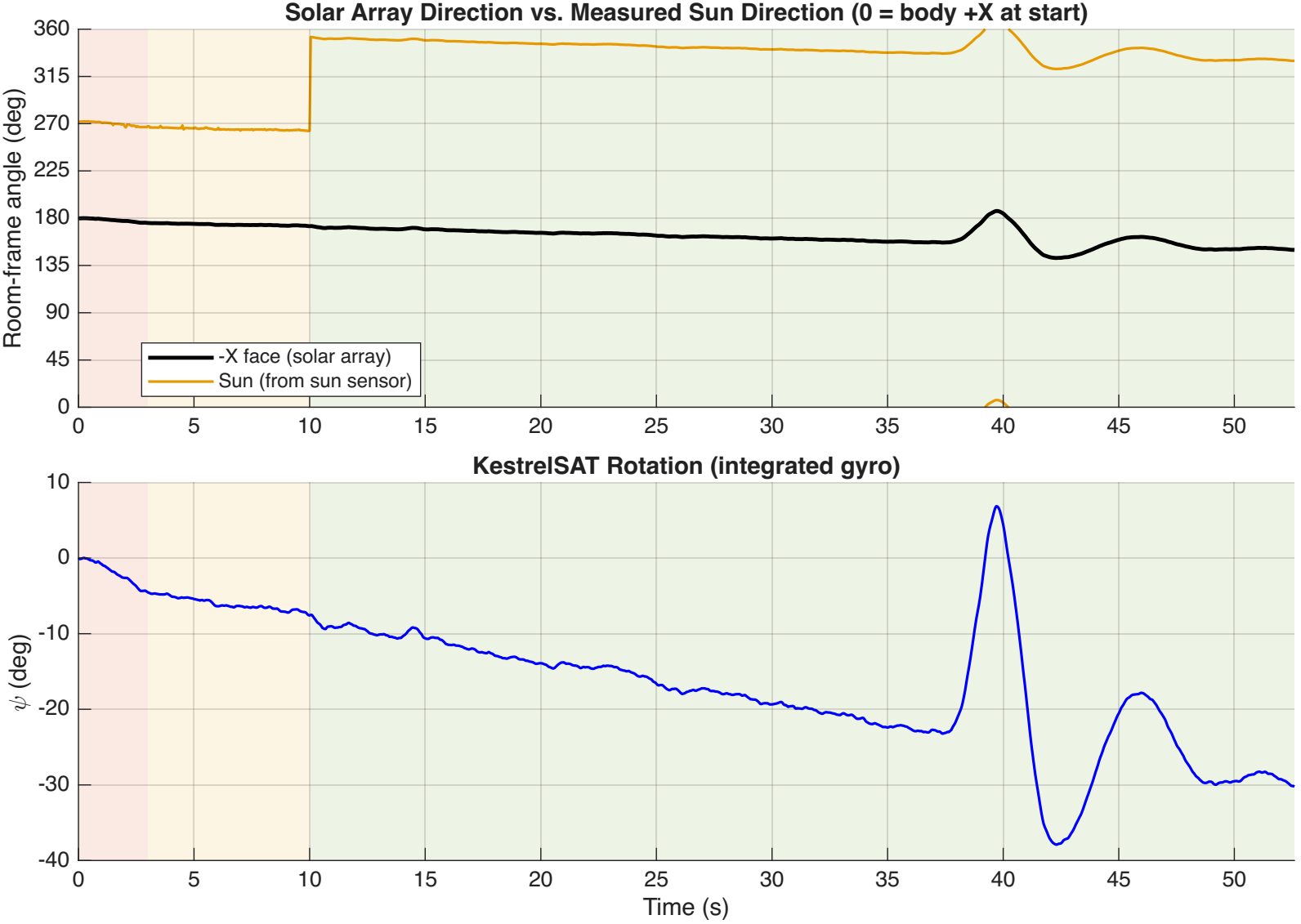

Z-Axis Body Rate (Gyro)
Body rate (deg/s)
Time (s)
SPINUP
HOLD
TRACK
30
20
10
0
-10
-20
-30
-40
0
5
10
15
20
25
30
35
40
45
50

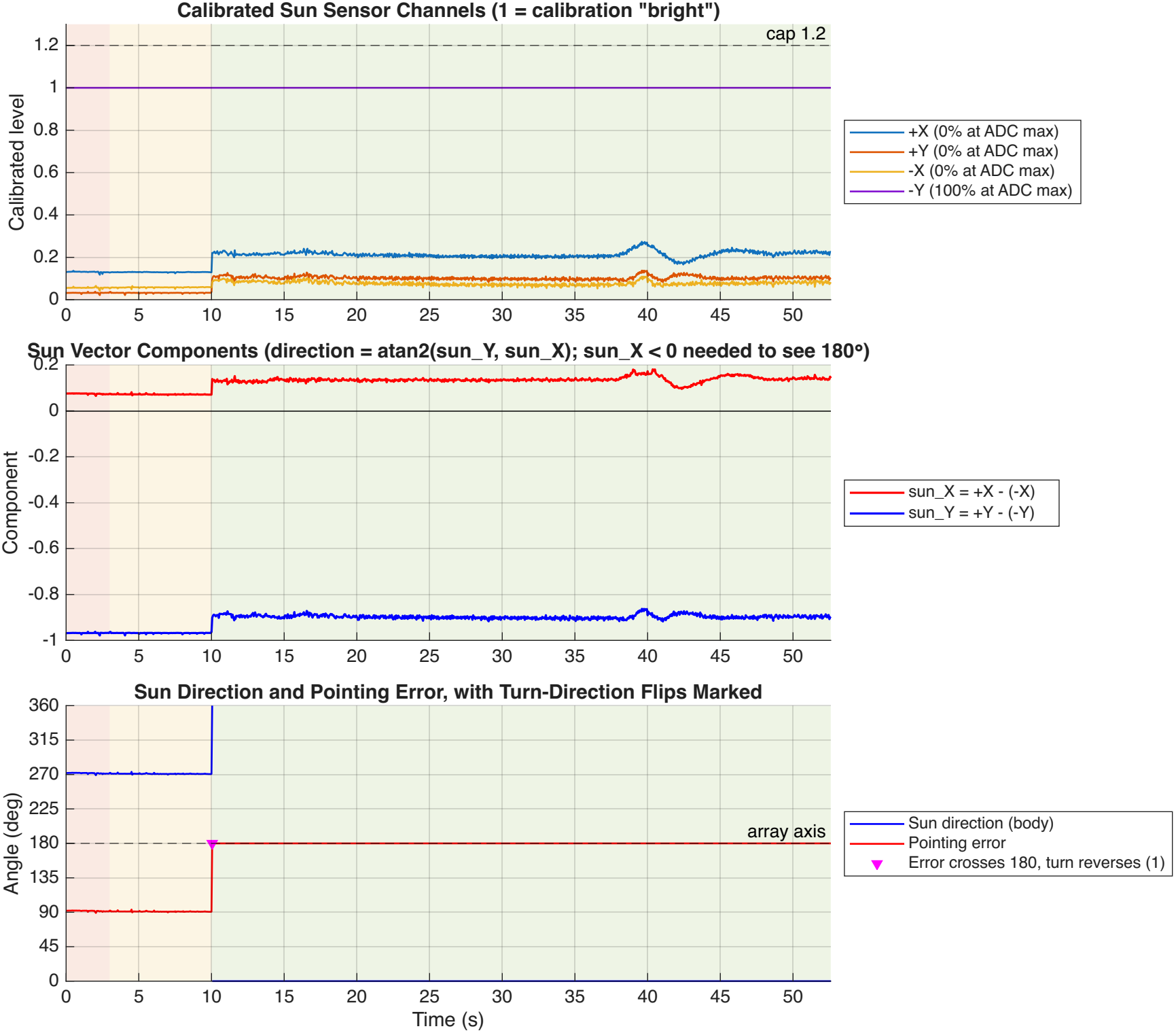

Reaction Wheel Speed
Wheel speed (RPM)
400
300
200
100
0
-100
0
5
10
15
20
25
30
35
40
45
50
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
400
200
0
-200
-400
-600
-800
Torque cmd
Saturated
Time (s)


Sun Sensor Photodiode Readings
ADC counts
ADC max
+X
+Y
-X
-Y
Sun Vector Magnitude
sun_mag
Time (s)


Magnetometer
magX
magY
|Bxy|
Field (μT)
Time (s)
Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)


Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
5
10
15
20
25
30
35
40
45
50
Time (s)
Sun direction
Mag heading atan2(Y,X)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
-X face (solar array)
Sun (from sun sensor)
0
5
10
15
20
25
30
35
40
45
50
KestrelSAT Rotation (integrated gyro)
ψ (deg)
10
0
-10
-20
-30
-40
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (0% at ADC max)
+Y (0% at ADC max)
-X (0% at ADC max)
-Y (100% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Error crosses 180, turn reverses (1)
Time (s)